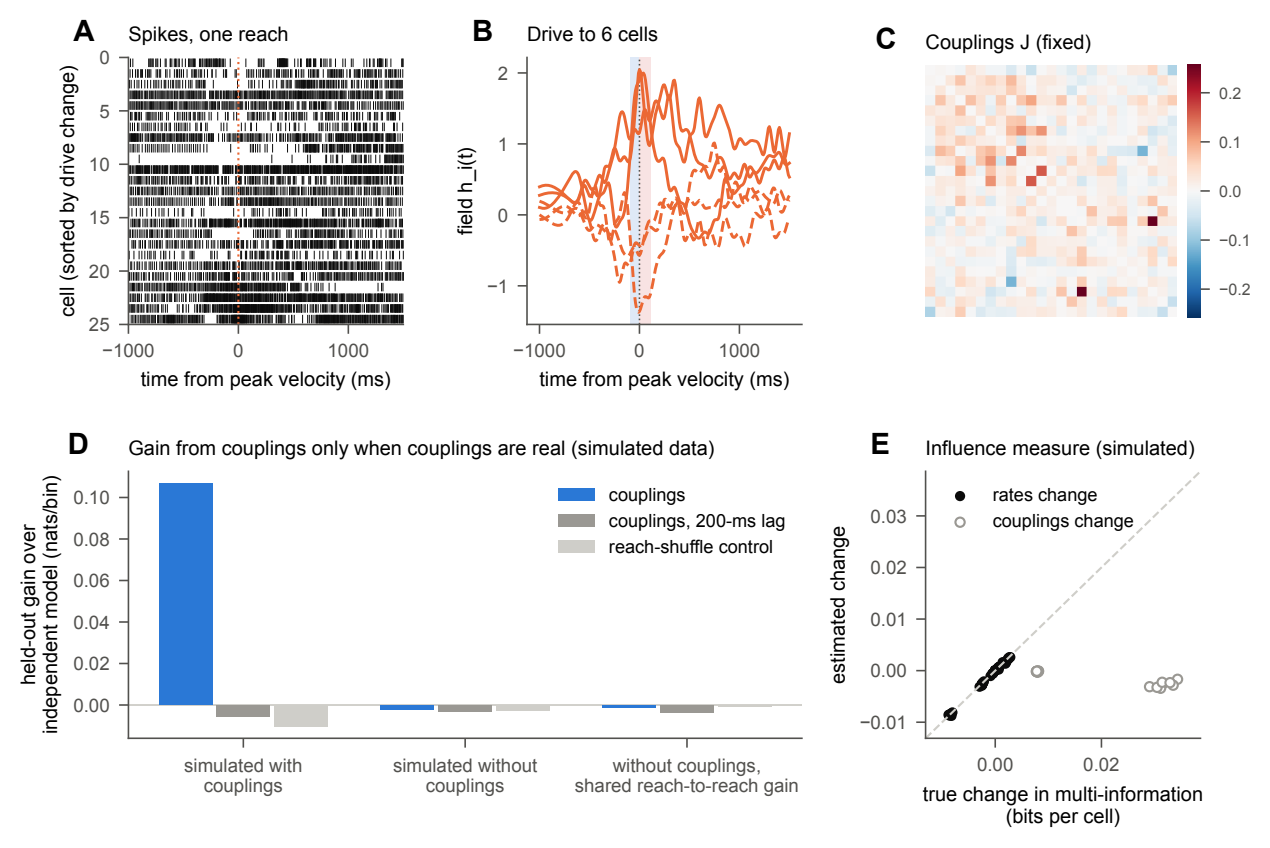
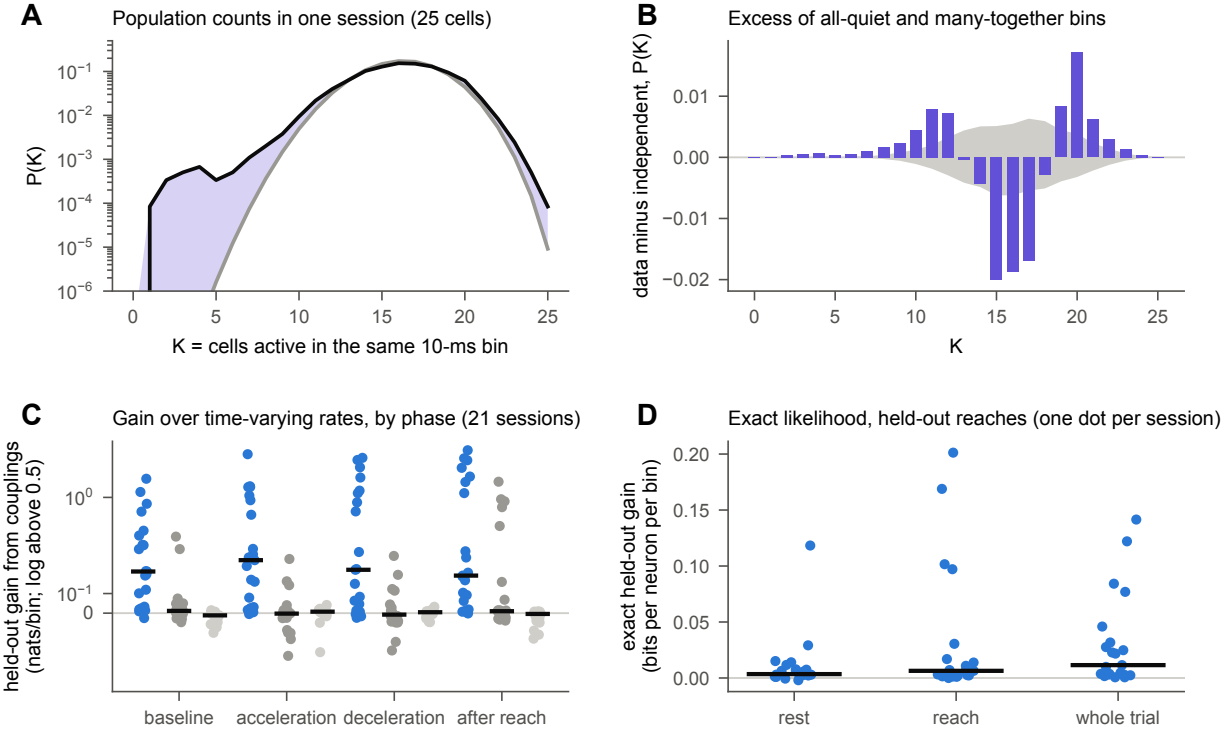
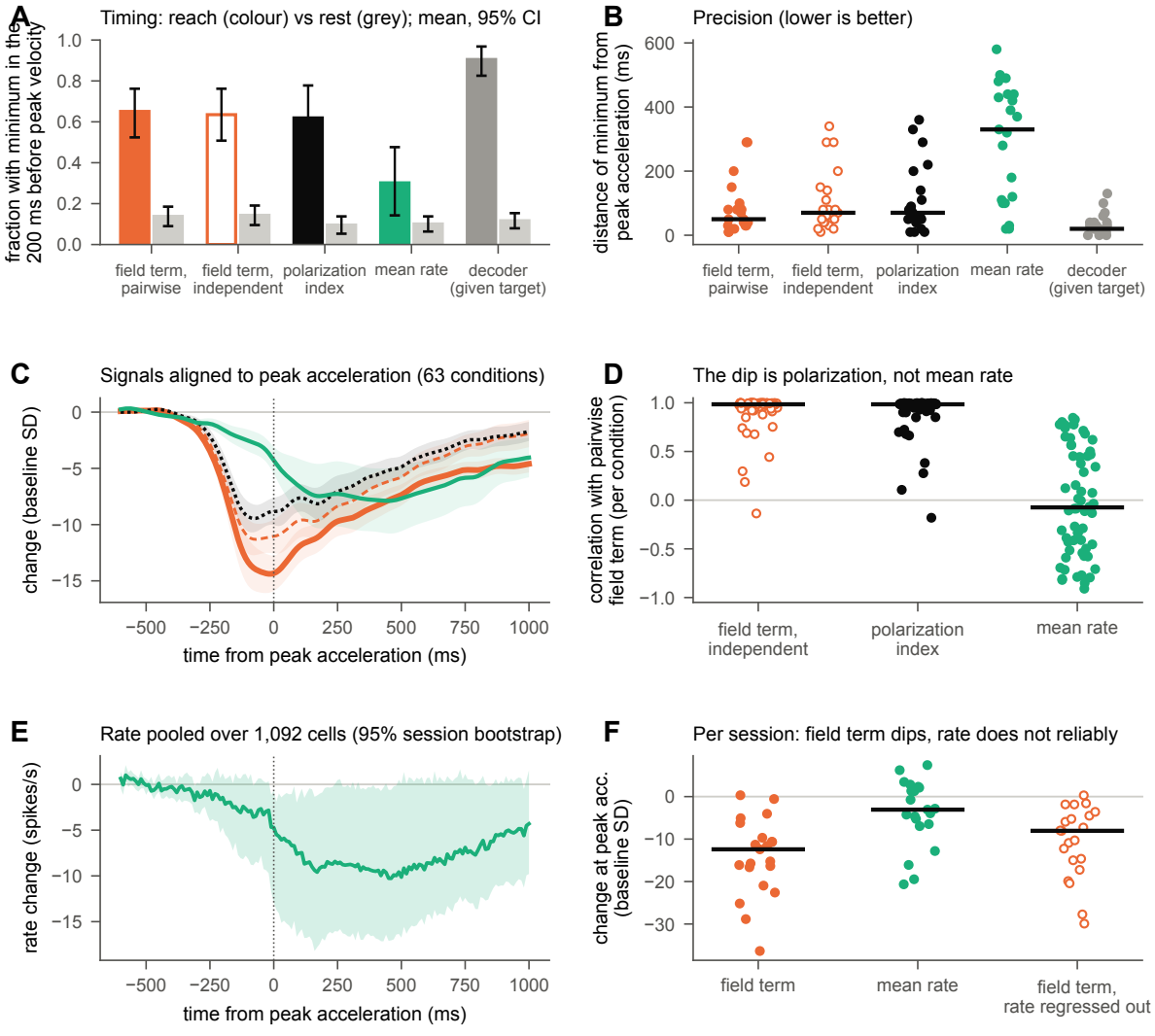
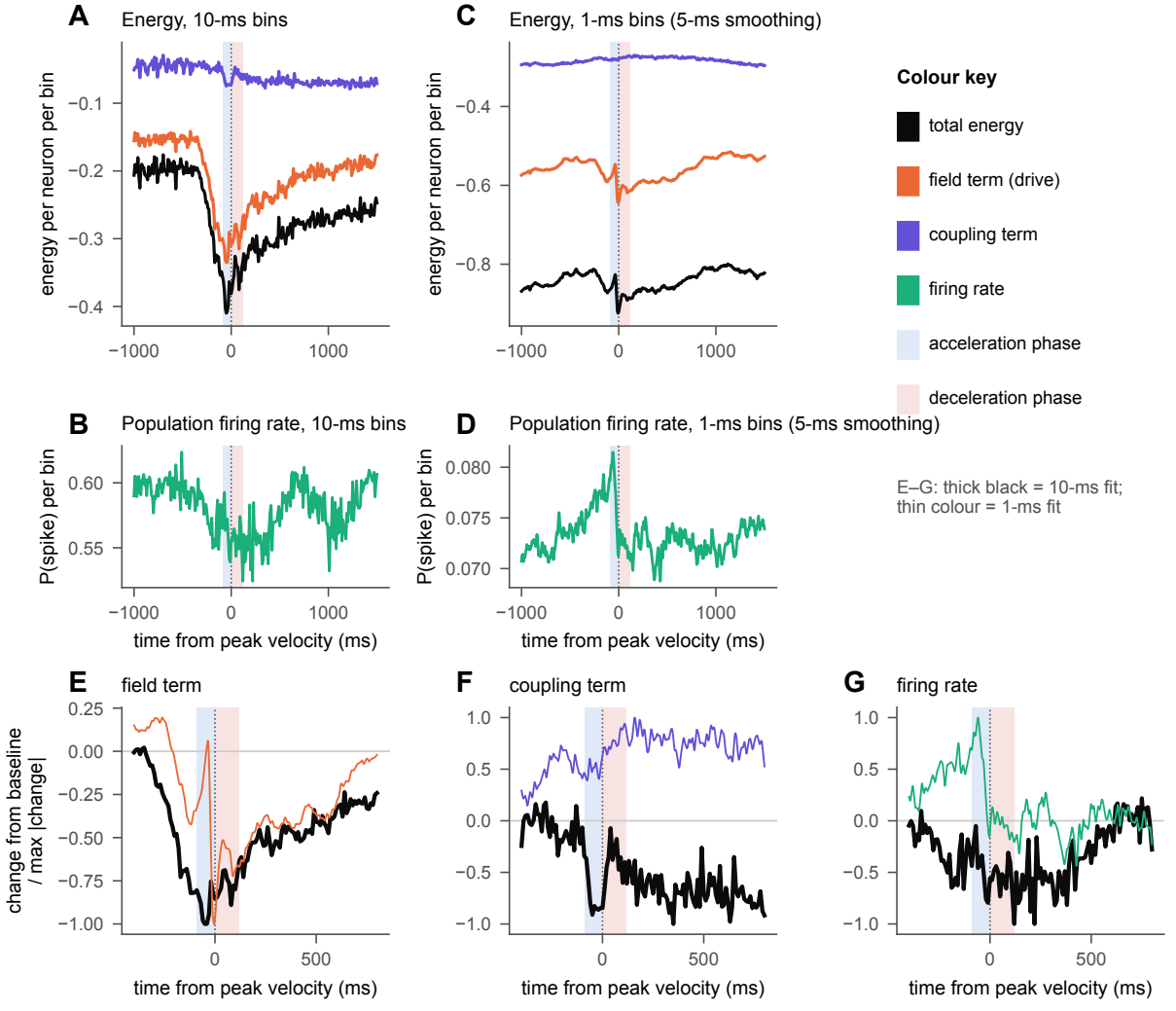
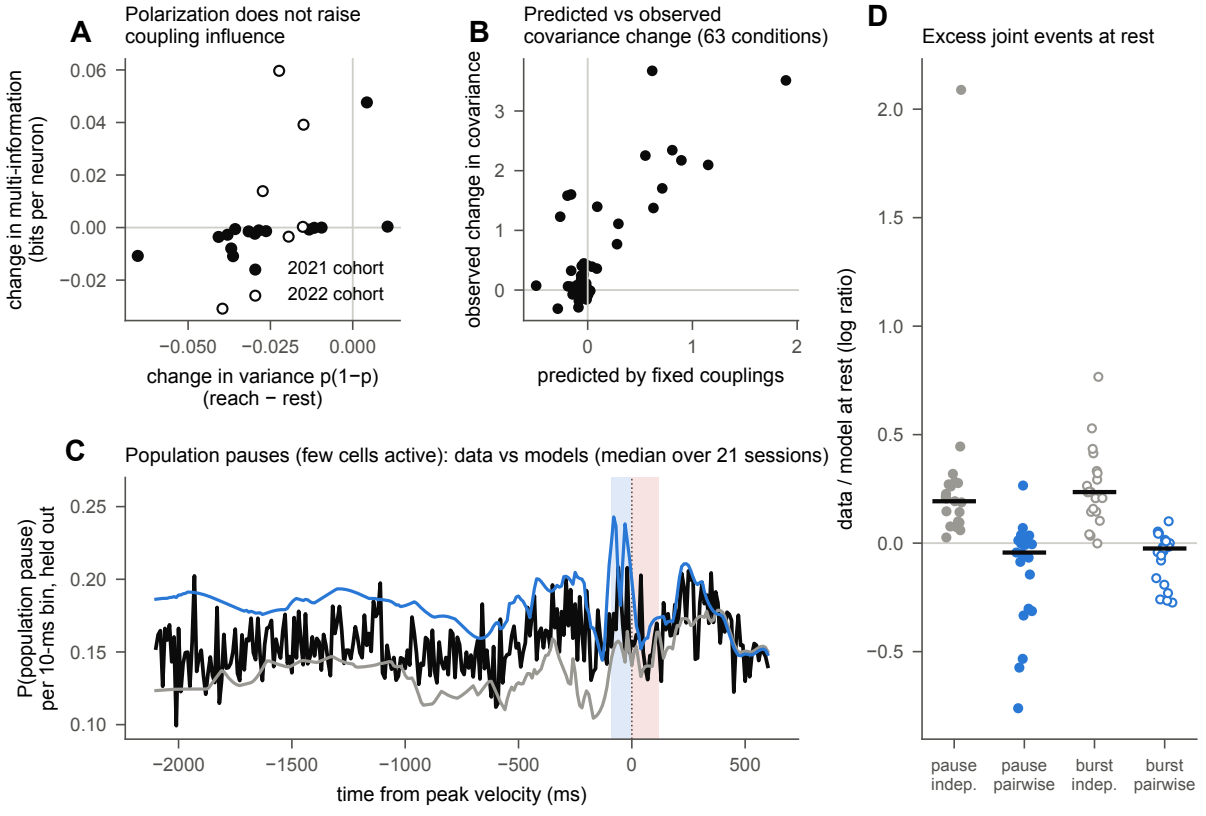
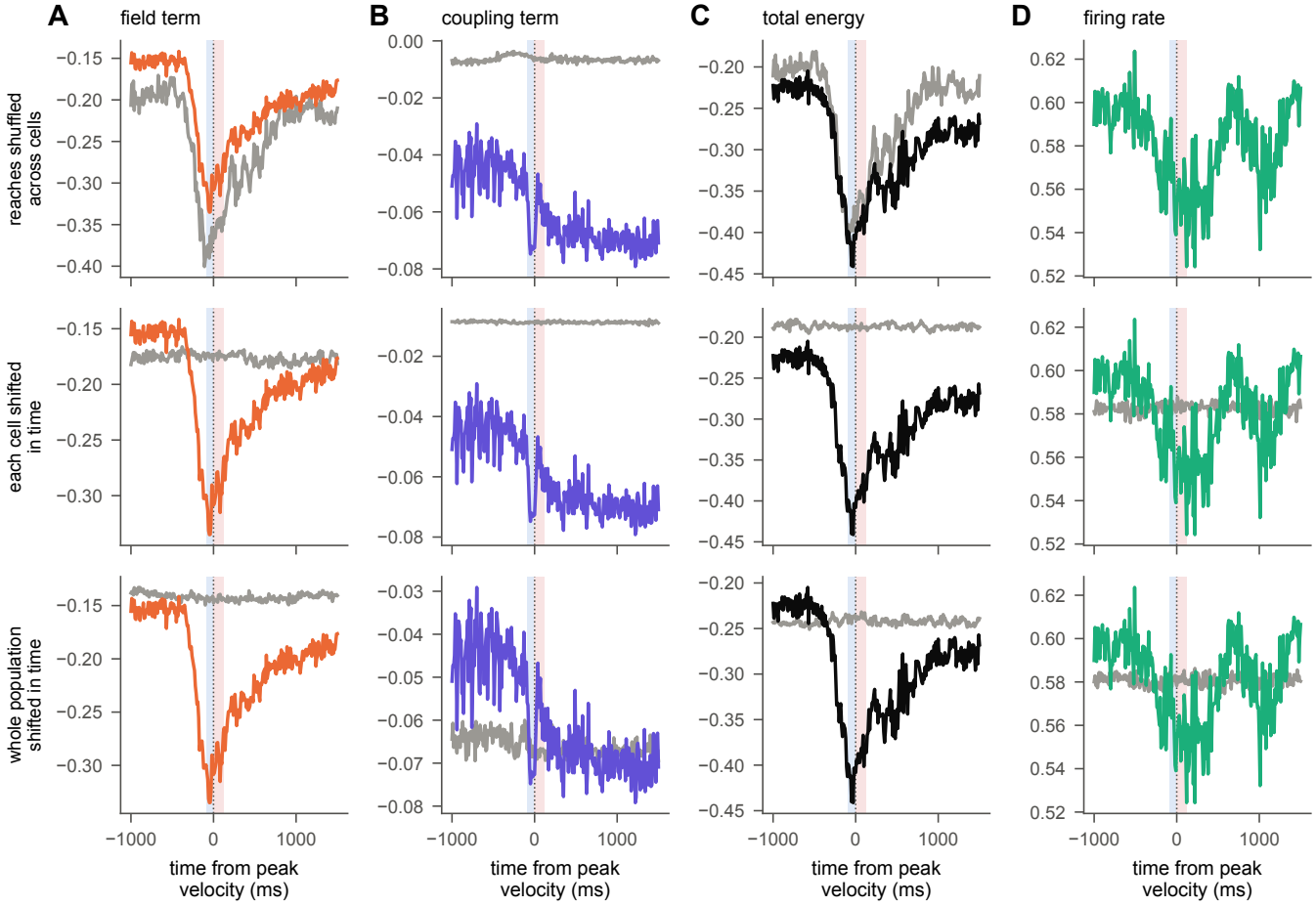

A
Spikes, one reach
cell (sorted by drive change)
0
5
10
15
20
25
−1000
0
1000
time from peak velocity (ms)
B
Drive to 6 cells
field h_i(t)
2
1
0
−1
−1000
0
1000
time from peak velocity (ms)
C
Couplings J (fixed)
0.2
0.1
0.0
−0.1
−0.2
D
Gain from couplings only when couplings are real (simulated data)
held-out gain over independent model (nats/bin)
0.10
0.08
0.06
0.04
0.02
0.00
couplings
couplings, 200-ms lag
reach-shuffle control
simulated with couplings
simulated without couplings
without couplings, shared reach-to-reach gain
E
Influence measure (simulated)
rates change
couplings change
estimated change
0.03
0.02
0.01
0.00
−0.01
0.00
0.02
true change in multi-information (bits per cell)

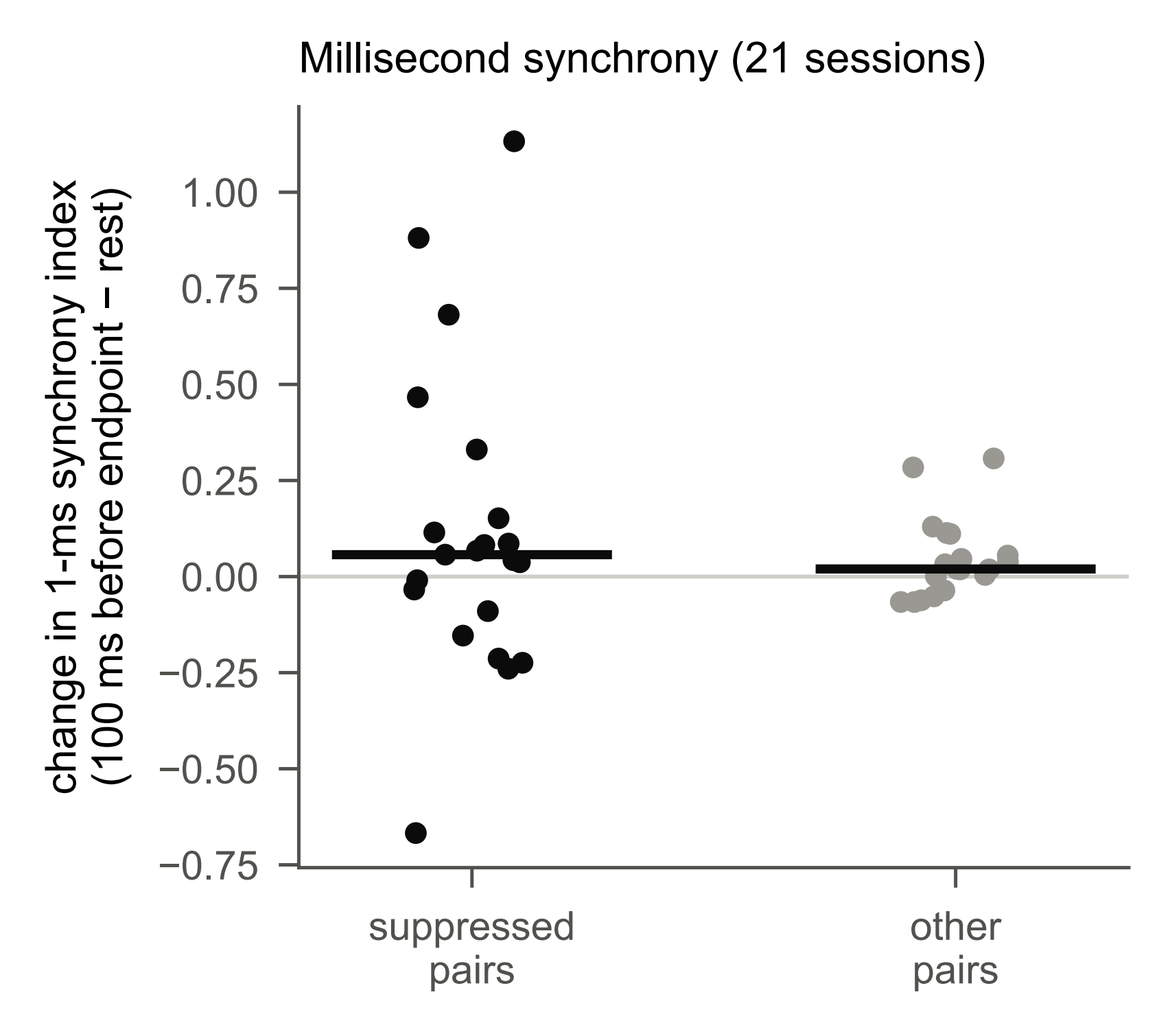

A
Population counts in one session (25 cells)
P(K)
K = cells active in the same 10-ms bin
B
Excess of all-quiet and many-together bins
data minus independent, P(K)
K
C
Gain over time-varying rates, by phase (21 sessions)
held-out gain from couplings
(nats/bin; log above 0.5)
baseline
acceleration
deceleration
after reach
D
Exact likelihood, held-out reaches (one dot per session)
exact held-out gain
(bits per neuron per bin)
rest
reach
whole trial


A
Timing: reach (colour) vs rest (grey); mean, 95% CI
fraction with minimum in the 200 ms before peak velocity
1.0
0.8
0.6
0.4
0.2
0.0
field term, pairwise
field term, independent
polarization index
mean rate
decoder (given target)
B
Precision (lower is better)
distance of minimum from peak acceleration (ms)
600
400
200
0
field term, pairwise
field term, independent
polarization index
mean rate
decoder (given target)
C
Signals aligned to peak acceleration (63 conditions)
change (baseline SD)
0
−5
−10
−15
−500
−250
0
250
500
750
1000
time from peak acceleration (ms)
D
The dip is polarization, not mean rate
correlation with pairwise field term (per condition)
1.0
0.5
0.0
−0.5
−1.0
field term, independent
polarization index
mean rate
E
Rate pooled over 1,092 cells (95% session bootstrap)
rate change (spikes/s)
0
−5
−10
−15
−500
−250
0
250
500
750
1000
time from peak acceleration (ms)
F
Per session: field term dips, rate does not reliably
change at peak acc. (baseline SD)
0
−10
−20
−30
field term
mean rate
field term, rate regressed out


A
Energy, 10-ms bins
energy per neuron per bin
−0.1
−0.2
−0.3
−0.4
−1000
0
1000
C
Energy, 1-ms bins (5-ms smoothing)
energy per neuron per bin
−0.4
−0.6
−0.8
−1000
0
1000
Colour key
total energy
field term (drive)
coupling term
firing rate
acceleration phase
deceleration phase
B
Population firing rate, 10-ms bins
P(spike) per bin
0.60
0.55
−1000
0
1000
time from peak velocity (ms)
D
Population firing rate, 1-ms bins (5-ms smoothing)
P(spike) per bin
0.080
0.075
0.070
−1000
0
1000
time from peak velocity (ms)
E–G: thick black = 10-ms fit;
thin colour = 1-ms fit
E
field term
change from baseline
/ max |change|
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
500
time from peak velocity (ms)
F
coupling term
1.0
0.5
0.0
−0.5
−1.0
0
500
time from peak velocity (ms)
G
firing rate
1.0
0.5
0.0
−0.5
−1.0
0
500
time from peak velocity (ms)


A
Polarization does not raise coupling influence
change in multi-information (bits per neuron)
0.06
0.04
0.02
0.00
−0.02
2021 cohort
2022 cohort
−0.050
−0.025
0.000
change in variance p(1−p) (reach − rest)
B
Predicted vs observed covariance change (63 conditions)
observed change in covariance
3
2
1
0
0
1
2
predicted by fixed couplings
C
Population pauses (few cells active): data vs models (median over 21 sessions)
P(population pause) per 10-ms bin, held out
0.25
0.20
0.15
0.10
−2000
−1500
−1000
−500
0
500
time from peak velocity (ms)
D
Excess joint events at rest
data / model at rest (log ratio)
2.0
1.5
1.0
0.5
0.0
−0.5
pause indep.
pause pairwise
burst indep.
burst pairwise


A
field term
B
coupling term
C
total energy
D
firing rate
reaches shuffled across cells
each cell shifted in time
whole population shifted in time
−0.15
−0.20
−0.25
−0.30
−0.35
−0.40
0.00
−0.02
−0.04
−0.06
−0.08
−0.03
−0.05
−0.07
−0.45
0.62
0.60
0.58
0.56
0.54
0.52
−1000
0
1000
time from peak velocity (ms)


Millisecond synchrony (21 sessions)
change in 1-ms synchrony index
(100 ms before endpoint – rest)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
suppressed
pairs
other
pairs